```
# Problem 1
n = int(input())
size = 2 * n - 1

for i in range(size):
    row_str = ""
    for j in range(size):

        dist_top = i
        dist_bottom = size - 1 - i
        dist_left = j
        dist_right = size - 1 - j

        min_dist = min(dist_top, dist_bottom, dist_left, dist_right)
        val = n - min_dist

        if val % 2 != 0:
            row_str += str(val)
        else:
            row_str += "*"
    print(row_str)
```

```
5
555555555
5*******5
5*33333*5
5*3***3*5
5*3*1*3*5
5*3***3*5
5*33333*5
5*******5
555555555
```

```
# Problem 2
rows = int(input())

for i in range(rows):
    row_str = " " * (rows - 1 - i)
    c = 1
    for j in range(i + 1):
        if c % 2 != 0:
          symbol = "#"
        else:
          symbol = "."
        if j == 0:
            row_str += symbol
        else:
            row_str += " " + symbol

        c = c * (i - j) // (j + 1)

    print(row_str)
```

Show hidden output

```
# Problem 3
x = int(input())
arr = []

while x != 1 and x not in arr:
    arr.append(x)
    a = x
    next_x = 0
    while a > 0:
        digit = a % 10
        next_x += digit * digit
        a //= 10
    x = next_x

print(f"Chain: {arr}")
print(f"Final x: {x}")
print(f"Replacements: {len(arr)}")
```

```
7
Chain: [7, 49, 97, 130, 10]
Final x: 1
Replacements: 5
```

```
# Problem 4
limit = int(input())
```

```
accepted = []
total_sum = 0
a, b = 1, 1
stopped_at = 0

while True:
    term = a
    a, b = b, a + b

    if term % 3 == 0:
        continue
    if term not in accepted:
      accepted.append(term)
    total_sum += term

    if total_sum > limit:
        stopped_at = term
        break

print(f"Accepted terms: {accepted}")
print(f"Sum: {total_sum}")
print(f"Count: {len(accepted)}")
print(f"Stopped at: {stopped_at}")
```

```
100
Accepted terms: [1, 2, 5, 8, 13, 34, 55]
Sum: 119
Count: 7
Stopped at: 55
```

```
# Problem 5
N = int(input())
P = int(input())

total_count = 0

for a in range(1, N):
    for b in range(a, N):
        c = 1
        while c * c < a * a + b * b:
            c += 1

        if c * c == a * a + b * b and a + b + c <= P:
            print(a, b, c)
            total_count += 1

print(f"Total: {total_count}")
```